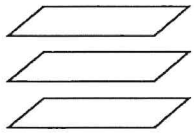
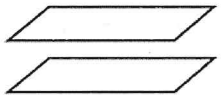
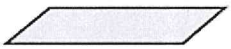
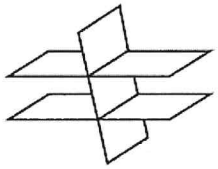
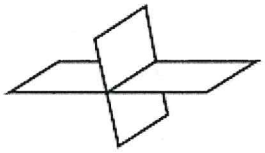
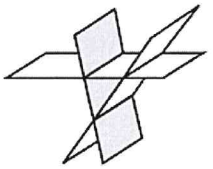
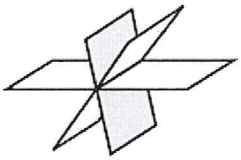
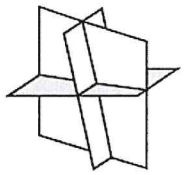
Intersection of Three Planes

Diagram	Description of Planes	Normal Vectors Comparison	# of Solution
	3 parallel planes	$\vec{n_1} \not\parallel \vec{n_2} \not\parallel \vec{n_3}$ (all parallel)	$\emptyset$
	2 coincident planes, the 3rd parallel	$\vec{n_1} \not\parallel \vec{n_2} \not\parallel \vec{n_3}$ (all parallel)	$\emptyset$
	3 coincident planes	$\vec{n_2} \not\parallel \vec{n_2} \not\parallel \vec{n_3}$ (all parallel)	Infinite # (plane)
	2 parallel and distinct planes the 3rd crossing	$\vec{n_1} \not\parallel \vec{n_2}$	$\emptyset$
	2 planes coincident the 3rd crossing	$\vec{n_1} \not\parallel \vec{n_2}$	Infinite # (Line)
	3 non-parallel planes but no intersection	normals are coplanar $\vec{n_1} \cdot (\vec{n_2} \times \vec{n_3}) = 0$	$\emptyset$
	3 planes all meet at one line	normals are coplanar $\vec{n_1} \cdot (\vec{n_2} \times \vec{n_3}) = 0$	Infinite # (line)
	3 planes intersect at one point.	normals are not parallel and not coplanar	1 (point)

Example 1: Determine the intersection of the planes and classify the planes.

① $3x+2y+4z=9$ $\vec{n_1}=(3,2,4)$
② $x-y+3z=-2$ $\vec{n_2}=(1,-1,3)$
③ $x+4y-3z=14$ $\vec{n_3}=(1,4,-3)$
} not //

$\vec{n_1}\cdot(\vec{n_2}\times\vec{n_3})=5$
∴ non-coplanar.
∴ 1 p.o.i.

Eliminate x (using ② & ③)

② $x-y+3z=-2$
⊖ ③ $x+4y-3z=14$
$-5y+6z=-16$ ④

Eliminate x (using ① & ②)

① $3x+2y+4z=9$
⊖ ② × 3 $3x-3y+9z=-6$
$5y-5z=15$
$y-z=3$ ⑤

Eliminate y (using ④ & ⑤)

④ $-5y+6z=-16$
⊕ ⑤ × 5 $5y-5z=15$
$z=-1$

sub $z=-1$ into ⑤
$y-(-1)=3$
$y=2$

sub into ②
$x-(z)+3(-1)=-2$
$x=3$

∴ P.O.I. $(3,2,-1)$

3 non-parallel planes intersecting @ 1 point

Example 2: Determine the intersection of the planes and classify the planes.

① $x+y+2z=-2$ $\vec{n_1}=(1,1,2)$
② $3x-y+14z=6$ $\vec{n_2}=(3,-1,14)$
③ $x+2y=-5$ $\vec{n_3}=(1,2,0)$
} not //

$\vec{n_1}\cdot(\vec{n_2}\times\vec{n_3})=0$
∴ coplanar.
∴ no solution or a line.

Eliminate x (using ① & ②)

① × 3 $3x+3y+6z=-6$
⊖ ② $3x-y+14z=6$
$4y-8z=-12$
$y-2z=-3$ ④

Eliminate x (using ① & ③)

① $x+y+2z=-2$
⊖ ③ $x+2y=-5$
$-y+2z=3$
$y-2z=-3$ ⑤

Eliminate y

④ $y-2z=-3$
⊖ ⑤ $y-2z=-3$
$0z=0$
infinite solution

∴ Let $z=t$

④ $y-2t=-3$
$y=-3+2t$

③ $x+2y=-5$
$x+2(-3+2t)=-5$
$x-6+4t=-5$
$x=1-4t$

∴ line: $\vec{r}=(1,-3,0)+t(-4,2,1), t\in\mathbb{R}$